

KARTHIK TATINENI

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EDUCATION

B.Tech in Electronics and Communication Engineering

2022 – 2026

Institute of Aeronautical Engineering, Hyderabad | CGPA: 7.2/10

SKILLS

Languages: Python, JavaScript, SQL

Web: FastAPI, Node.js, REST APIs, React, HTML, CSS, Tailwind CSS

AI/ML: YOLO, CNN, RAG, MCP, AI Agents, Embedding

DevOps/Cloud: Docker, Kubernetes, CI/CD, AWS (EC2, S3, IAM), Azure (Container Apps, Container Registry), Azure DevOps, Azure Foundry, Cloudflare, Grafana

Databases: Firebase, Supabase, MongoDB, MySQL, PostgreSQL, Redis

Tools: Git, GitHub, JupyterLab, Anaconda, Postman, Vercel, Umami, Claude Code CLI, Copilot CLI, Hugging Face, Ollama, Nvidia NIM

EXPERIENCE

Freelance Software Developer

Jan 2025 – Present

- Built **KLIVORA Fashion Hub**, a production-ready Next.js e-commerce platform with Firebase, Supabase, AI concierge, Three.js, secure authentication, checkout, inventory management, and order tracking. [\[Live\]](#)
- Built **Touchwood Interiors CMS**, a production-ready Next.js platform with Sanity CMS, Supabase, AI support chatbot, secure quotations, interactive 3D visualization, and architect portfolios. [\[Live\]](#)

PROJECTS

DisasterMind AI - Multi-Agent | *FastAPI, NVIDIA NIM, PostgreSQL, pgvector, RAG, Embedding* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Built a multi-agent RAG system using NVIDIA NIM LLMs, vector embeddings, and pgvector similarity search over PostgreSQL to power grounded, explainable disaster response reasoning across coordinator, risk, logistics, and evacuation agents.
- Engineered an async FastAPI backend with tool-calling agent orchestration, retrieving 12,000+ real-time records via RAG to reduce hallucination and power a Next.js/TypeScript decision-support dashboard.

Distributed College Event Management System | *React, Firebase, Docker, Nginx, Vercel, AWS* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Deployed a high-availability event platform handling 500+ live registrations with zero downtime across concurrent admin and participant sessions.
- Architected a scalable backend with microservices, Nginx load balancing, and Cloudflare CDN, achieving low response times under peak load.

Real-Time Helmet & Triple-Ride Detection | *Python, YOLOv3, CNN, TensorFlow, PyTorch, OpenCV* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Built a computer vision pipeline detecting helmet violations and triple-riding with 90%+ accuracy on test data; optimized to 25+ FPS for real-time traffic surveillance.
- Fine-tuned YOLOv3 on a custom-labeled dataset and implemented automated violation detection for law enforcement alert systems.

PUBLICATIONS

T. Karthik Sai et al., **Helmet Violation and Triple-Ride Detection Using YOLOv3 and CNN**, ICMCER 2026 (Scopus Indexed) | [Certificate](#)

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals | [Certificate](#)
- MySQL & Database Management: Create, Manage & Query Databases | [Certificate](#)
- GenAI For Image & Video Creation | [Certificate](#)
- HTML & CSS: The Complete Web Development Guide | [Certificate](#)
- Microsoft AI Skills Fest 2026 | [Credly Badge](#)

EXTRA-CURRICULAR

- Hosted and organised the official **IARE Consortium** event for 1,000+ participants; designed and developed the end-to-end registration platform.
- **Core Member and Logistics Head**, NSS – led community logistics for college and district level programs.